The Ethics of an Aesthetics of Wonder for Planetary Care

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ABSTRACT

In design, the expression of values through form and experience is often referred to as *aesthetics*. In response to the ecological crisis, we explore wonder as a design aesthetics for planetary care. This exploration occurs through the presentation and critique of three seasonal activities with children in Sweden, aimed at inspiring local ecological curiosity: *Appreciating the Wind*, *Looking for Frog Eggs*, and *Making Natural Perfumes*. From these activities, we reflect upon the ethics of an aesthetics of wonder. This includes addressing the prioritization of human experience and the commodification of nature in design. We offer three contributions: defining and positioning wonder as a possible design aesthetics for planetary care, presenting three seasonal activities that situate wonder as an aesthetics in practice, and offering reflections on the ethics of wonder based on a critique of the activities.

Keywords

children
more-than-human
seasonal design
place-based
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Design can be understood as proposing intentional change through purposeful additions to the real world (Nelson & Stolterman, 2012). This means that it relies on values and principles to guide what might be desirable, or "better", outcomes. The resulting expression of these principles is often referred to as *aesthetics*. For example, aesthetics might be expressed through physical shape, interactive behavior, or visual composition, and they are important in conveying an intended use, impression, or experience. In giving form to notions of "better" and in defining how a person might relate to an outcome, aesthetics is also a matter of ethics.

In this article, we explore wonder as a design aesthetics for planetary care and reflect upon the ethics of how it might affect human and more-than-human ways of living and being "as well as possible" (Tronto, 1993, p. 103). This exploration responds to questions concerning aesthetics for planetary co-habitability and is situated within a growing body of research on the ethical complexities of more-than-human aesthetics in design (Hauser et al., 2023; Ikeya et al., 2025; Sehgal & Wilkie, 2024).

We explore wonder through the presentation and critique of three seasonal activities with children, aimed at inspiring local ecological curiosity, care, and coexistence. Seasonal activities are everyday events for people of all ages to learn about environmental temporalities, outdoor habitats, and multispecies survival through observing, experiencing, and making (e.g., Stirling, 2021). We situate possible "learning" from such activities not as the gath-

ering of facts or a prescribed knowledge outcome, but instead, as the ongoing thinking and feeling encountered before, during, and after an activity. More simply, the intended experience is to ask questions. This situates the aesthetic intent as that of wonder, which, drawing upon MacLure (2013), we define as awe suspended between knowing and not knowing. In response to planetary concerns and within calls for new ways of designing that express different logics and values (Ávila, 2022; Escobar, 2018), we explore the aesthetics of wonder as a matter of care (Puig de la Bellacasa, 2017), with the potential to transform ecological relations, yet fraught with ethical tensions in defining ways of living and relating.

In the following, we first motivate wonder as a desired aesthetics for planetary care. Then, we ground a focus on early childhood (i.e., birth to eight years old) interactions with local outdoor ecologies, and more specifically, the doing of seasonal activities to explore wonder in practice-based design research. Next, we describe our larger project on seasonal activities and detail three activities with young children in Sweden: *Appreciating the Wind*, *Looking for Frog Eggs*, and *Making Natural Perfumes*. This detailing encompasses the intent of each activity as depicted in the Swedish book *Året runt med Pettson och Findus* (*All Year Round with Pettson and Findus*) (Nordqvist et al., 2023), the practical doing of the activity, and how wonder was expressed.

Notably, this interpretation of expression is provided by the first author as the organizer, a participant, and a parent in the three activities. Lastly, we reflect upon the ethics of wonder, which includes addressing the prioritization of human experience and the commodification of nature. From this, we propose the following contributions: the positioning of wonder as a design aesthetic, three seasonal activities that situate wonder in practice, and reflections on the ethics of wonder based on a critique of the three activities. We offer these contributions to designers who scaffold encounters between people of all ages and local ecologies. Scaffolding encounters includes the staging of more-than-human interventions during the setup of a design project and reflecting on unfolding experiences of use with a designed product.

Why Wonder for Planetary Care?

In writing about the potential of *wonder* in qualitative research, MacLure describes it as awe along a threshold between knowing and not knowing (2013). Through surprise, further thought is animated, and questions are provoked. Drawing upon Daston and Park (1998), MacLure further positions wonder as a "cognitive passion" that is as much about thinking about knowledge as it is about *feeling* curiosity, fascination, horror, or disgust in the gut or through a heartbeat. That is, it is material and somatic, not necessarily safe

or positive, and emerges through an entangled "capacity to affect and to be affected" (MacLure, 2013, p. 229). Stengers similarly defines wonder as being surprised and entertaining questions, yet also expands upon its relationality by describing it as a double operation, whereby being affected not only illuminates what is not known, but also calls into question what is known (Stengers, 2011, p. 374). This adds unknowing to the threshold within which wonder lies, emphasizing possibilities of disorder in welcoming what might complicate enduring principles and values that configure modes of relationality.

These perspectives ground wonder as a potential design aesthetics for planetary care. We position aesthetics in design practice as the judgments that guide decisions during a process and the expression and experience of an outcome. For example, simplicity as an aesthetics in industrial design is based on principles of honesty and ease of use, which are often expressed through low visual complexity and the experience of efficiency (Hauser et al., 2023). For us, wonder as an aesthetics is based on principles of relationality and affection, which are expressed through the staging of bodily encounters between people and more-than-human worlds and through the experience of questions.

In a design process, wonder resists prescribing what questions might or should be asked, and how their asking might be experienced or expressed. In relation to the defuturing of particular people and more-than-human ways of living through design (Fry, 2020), it aligns with practices of noticing that strengthen or unmake relationships (Liu et al., 2019; Poikolainen Rosén et al., 2024), and with regenerative thinking that supports environmental and cultural flourishing without a future orientation toward innovation and progress (Nicenboim et al., 2025). As a design outcome experienced through questions, wonder aims to stay with what is not known to give way to new socio-natural conceptions, and to stay with what might be unknown to disrupt extractivist logics. More simply, it invites an ongoing questioning of coexistence.

Early Childhood Planetary Care

We explore wonder as an aesthetics for planetary care in design practice through the doing of seasonal activities with children. Theoretically, we draw upon posthumanism to orient ourselves toward children as citizens and caretakers of the Earth (Hickey-Moody et al., 2021). The notion of posthuman children aims to reconfigure children as capable and impactful, yet often excluded, participants in efforts toward "better" ways of living (Murris, 2016). In design research, a growing body of work recognizes that, as inheritors of the climate crisis, children deserve the opportunity to assert their own ways

of relating to planetary entanglements now and to develop ecological sensitivities for the future (Anggarendra & Brereton, 2016). These works notably depart from a rich and important history of research on children outdoors, such as the design of playgrounds (e.g., Mårtensson et al., 2025), and explore how the scaffolding of multispecies play and ecological noticing might participate in broader efforts to reimagine design (Fry & Nocek, 2020). Building upon these efforts, we further orient towards early childhood (i.e., birth to eight years old) as a group often excluded from cultivating practices of ecological care (Vella et al., 2023) and draw upon temporality and more-than-human transformations within children's literature as a resource (Österlund et al., 2024).

In this article, seasonal activities are unskilled, creative pursuits during which functional or decorative objects are made by hand using simple tools. Additionally, they are temporally and materially scaffolded by local seasons to learn about and joyfully connect with nearby ecosystems (Stirling, 2021). In design research, this temporal focus aligns with the Seasonal Designing Framework through a reliance on "the potential of seasons as a resource to guide the design process and its outcomes" (Keune & Ståhl, 2025, p. 2). According to the authors, the potential of the seasons includes the availability of place-based elements (e.g., water, fire, earth, and wind) and the variability of environmental expressions (e.g., hail, snow, ice, rain, and sunshine).

We situate seasonal activities as design events (Jönsson, 2015) within the relational ontologies of children (Hickey-Moody et al., 2021). This means that the things between the child and the material world are in constant intra-action with the becoming of subjects and the assemblages to which they belong (Barad, 2003), and this includes not only physical objects resulting from activities, but also the preparations, encounters, and collaborations before, during, and after these activities with people of all ages. In this way, seasonal activities hold the potential to serve as a significant site of minor activism, in which "tiny changes are the beginning of much larger shifts" (Hickey-Moody et al., 2021, p. 34), such as children connecting to local outdoor ecologies through collaboratively and inventively engaging with materials, beings, and each other.

In summary, the inclusion of children is relevant to design practice and planetary care because it foregrounds possibilities for supporting the long-term becoming of place-based relations through material encounters and collective approaches that are temporally scaffolded by seasons. This aims to radically shift design practice from individualistic values, immediate gratification, and definitive interactions toward an emphasis on relationality, temporality, and wonder.

THREE SEASONAL ACTIVITIES WITH CHILDREN

In September 2023, the first author (FA) began doing seasonal activities with her partner and two children (both under eight years old) from the Swedish book *Året runt med Pettson och Findus* (Nordqvist et al., 2023). This book was chosen for the following reasons: first, it encompasses a full calendar year of activities (i.e., January-December) that are based on generalized seasons and natural materials from the FA's country of residence (Sweden). This situates the activities as place-based (Vanni & Crosby, 2023). Second, the activities are presented textually and illustratively through popular Swedish children's book characters, Pettson and Findus. This grounds the proposition and narration of the activities in the everyday lifeworld of the FA's family, and highlights a theoretical commitment to naturecultures (Haraway, 2003), which is furthered through the playful multispecies relationship between the characters (i.e., an older man and a cat). Third, the introduction of the book emphasizes that the activities are not just for children but rather for all ages and for doing together. This aligns with theoretical commitments to the relationality of children and foregrounds intergenerational collaboration.

Over the course of eleven months, forty activities were completed in or near her home in Sweden to situate engagement with local materials and outdoor ecologies, and with her family, neighbors, and friends. The FA was the organizer and a participant in all activities, which included the involvement of her family in thirty-four activities. Participation extended beyond the FA and her family to include children and adults from the neighborhood. Four of the activities included two to seven additional children, three of which are the focus of this paper. Formal ethical approval was not required since sensitive data was not collected, yet informed consent was obtained from all participants, including the FA's family.

The overarching organizing, doing, and subsequent critique of the activities methodologically foregrounds first-person reflexivity aimed at recontextualizing, rather than removing, human action in relation to environmental concerns (Helms, 2023; Nicenboim et al., 2023; Smith, 2019), and calls attention to the possibility of seasonal interventions as personal and collective artistic investigations between an author and participants (Keune & Ståhl, 2025). Amid the prevalence of the FA's role and multiple, differing configurations of participants, "we" is used to refer to all the authors of the article for clarity.

Children's interests took priority during all activities. Each was treated as a regular play date with friends of the FA's children, which included the possibility of doing a material experiment or crafting an artifact. During and after each activity, the FA took photos and notes from her perspective

as an organizer, participant, and parent. This forms the basis of our presentation and critique. This documentation centers on the FA's experience and her perspective regarding the impressions and expressions of wonder. Not relying upon verbal communication by children to validate the presence of wonder, and allowing for embodied making to articulate possibilities of experience, were ethical choices situated within adult-child power relations (Wiedenman et al., 2023). Thus, in drawing upon scholarship on posthuman children (Murris, 2016), we aim to be transparent regarding the FA's influence over the activities and the interpretive analysis of them.

For each activity, we first present its intent in relation to planetary care and the expected outcome as positioned by the book. We then describe the FA's intentions as a staging of relations to be explored through those expected outcomes. This includes motivating differences from the book and describing who participated. Next, we describe events before, during, and after seasonal activities to foreground how an aesthetics of wonder was experienced, as interpreted by the FA.

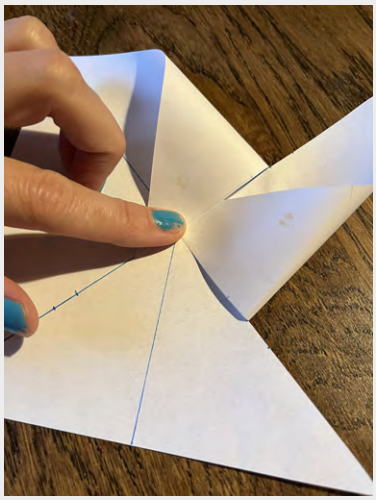
Appreciating the Wind

Appreciating the Wind took place in October, a month characterized by changes in the weather. The book suggests activities to observe and measure the local atmosphere. Desired outcomes include watching tree branches sway with the wind, observing moss spores when humidity is high, making devices to spin in the wind, and transforming a plastic bottle into a rain meter. The interpreted intent of the outcomes is to relate environmental behaviors to systems and scales beyond human perception, such as the coolness of a breeze or the movement of a tree during a change in air pressure.

When carrying out the activity, the FA focused on appreciating the wind through the making of pinwheels, which are toys with vanes on a stick that spin in the wind. This focus aimed to explore the functional dynamics and force of the wind through an abstract and playful representation. That is, rather than focusing on how the wind feels on the human body, the intent was to explore how the wind, as it is felt, works through directionality and movement. This activity took place twice: first between the FA and her eldest child, and again with two additional children (siblings) and a parent.

The first occasion of making pinwheels took place on a windy weekday morning before preschool. The FA and her eldest child followed the instructions in the book to measure, color, cut, and fold a piece of paper into a three-dimensional form. On the walk to preschool, the FA's child held the pinwheel as it spun in the wind. On a subsequent, non-windy day, the child discovered that she could run to make it spin. A few weeks later, the FA invited

Figure 4: Process and outcomes of *Appreciating the Wind*: **a**) Measuring and folding paper to make pinwheels; **b**) Paper pinwheel spinning in the wind; **c**) Running to make a pinwheel spin; **d**) Setup of the second occasion of the seasonal activity with additional children. Photographs: Karey Helms, 2023.



a



b



c



d

two friends of her older child (siblings) and a parent to make pinwheels after preschool. During the activity, one child did not engage at all and instead played with other toys, and the other invited child remarked to his parent that the new teacher at preschool "doesn't let us play anymore either," and then left the table to play in another room.

The staging of relations between people and the wind was mediated by the pinwheel as an outcome and the process of its construction. During the first iteration, the FA was surprised by the difficulty of the pinwheel construction, which unpacked the possibility of motion without explaining the dynamics of wind. Yet, this unexpected admiration was not shared by all participants, and there was notable annoyance from a child during the second iteration of the activity. The use of a pinwheel on the first windy morning highlighted the absence of the wind on another morning. Yet, its effects on the pinwheel were easily recreated by the body of the FA's child through running. What, then, does a pinwheel capture, measure, or represent if the wind is so easily simulated by a person? Or is the wind as a force ever actually absent, or instead always present through its latent possibility? This possibility goes beyond illuminating what is not known about the wind and instead calls into question what is known about embodied encounters between people and the wind.

Looking for Frog Eggs

Looking for Frog Eggs took place in the month of April, a time associated with the observance of new life. The book suggests catching eggs and putting them into a jar of water to watch them transform into tadpoles and frogs. The desired outcomes are creating an aquarium habitat; crafting a fishing net from a metal hanger, mesh curtain, and wooden rod; and looking for frog eggs at a pond to populate the aquarium. The interpreted intent of the activity is to show how everyday household objects can be used to observe signs and cycles of life that might be hidden underwater and temporally misaligned with how people experience reproduction.

When carrying out the activity, the FA decided not to make the aquariums and instead to emphasize searching for frog eggs and other water creatures. This was because she was not sure how frog eggs could be responsibly supported in an artificial environment. Two friends of the FA's eldest child and one parent each were invited to join the FA's two children and partner for a trip to look for frog eggs on a Sunday afternoon at a nearby lake and pond. In preparation for the activity, the FA made three fishing nets for the older children from bamboo rods, cable ties, tulle, twine, and super glue.

The day of the activity was chilly and rainy. The group of four children and four adults left together from the FA's home to walk toward two areas

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Figure 2: Process and outcomes of Looking for Frog Eggs: **a**) Fishing nets made by the first author; **b**) Walking along the periphery of the pond; **c**) Scooping for eggs; **d**) A dead frog caught in a net. Photographs: Karey Helms, 2024.



a



b



c



d

where the FA speculated frog eggs might be found, based on seeing frogs mating the year before and on recently seeing dead frogs on a dirt path. The first spot was an open bay of a large lake that was extremely windy, so they instead walked to a pond sheltered by land and trees. On the way, the children found dead frogs on the path and by a storm drain, and stopped to examine their bodies. At the pond, while parents scouted the periphery for signs of eggs, the children stood on broken reeds to reach the deeper water and scoop for eggs. At differing moments, the children caught leaves: a bloated, dead frog, spiders, algae, a crawfish or beetle, possible tadpoles, and two mating frogs.

The active searching for, but failure to find, frog eggs was contrasted by encounters with death and other unidentifiable creatures, which revealed a threshold between knowing and not knowing about the location of the eggs. The encounters materially supported the possibility of eggs: before death, there must be life, and life is possible here and now. Yet, why was so much death encountered? Is this seasonally typical for frogs or symptomatic of something else? Searching also disordered boundaries between bodies within a shared ecology. This was evidenced by adults walking around the periphery of the pond and children trying to push beyond the edge of the land as far as possible by stepping on broken reeds. Boundaries transformed from scenic appreciation to felt frustration through the questioning of human access and capability. If frog eggs should be found but are not, what else cannot be sensed or found by human bodies? Yet the life that was found, such as spiders and algae, evoked an admiration of survival and a celebration of patience.

Making Natural Perfumes

Making Natural Perfumes took place in the month of May, a time associated with the thriving of plants and insects. The book suggests activities to support, sense, and appreciate flourishing. For example, desired outcomes include building an insect hotel, cultivating an herb garden, arranging a plant exchange, and capturing the aroma of flowers and spices in jars to play a scent memory game. The interpreted intent of the outcomes is to emphasize that life does not thrive in isolation but rather requires habitats, resources, and sensitivities to the specificities that support growth.

When carrying out the activity, the FA focused on the aromatic appreciation of plants by making natural perfumes, which are fragrant liquids crafted from the scents of non-synthetic materials. This was intended to emphasize scent as a possible sign of flourishing, and the mixing of scents as an exploration of what might flourish well together. The FA organized a perfume-making workshop in a community yard for her eldest child, two

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Figure 3: Process and outcomes of Making Natural Perfumes: **a**) Making essential oils from orange rinds; **b**) Setup of the perfume workshop; **c**) A resulting natural perfume; **d**) Mold growing inside a perfume bottle. Photographs: Karey Helms, 2024.



a



b



c



d

friends, and their parents. Three additional neighborhood children and two parents (one accompanying his two children) also joined after expressing interest while playing nearby.

A couple of days before the workshop, the FA and her children made essential oils from dried lavender, dried roses, orange rinds, and star anise by heating the materials in water on the stove and then mixing the results with flavored vodka to enhance the scents and prevent rotting. The scents were stored in glass tubes. The following materials were provided on the day of the workshop: homemade essential oils and water; fresh fruits and spices; dried fruits and flowers; a grater, a mortar and pestle; scissors, funnel, and pipettes; glass bottles; glitter; and colored pipe cleaners. When offered a bottle, children were instructed to explore the materials through smell and touch, and to fill and decorate their bottle with the help of parents. All children, including the FA's eldest child, took home their customized natural perfume. Despite the presence of alcohol, mold eventually grew inside, and the FA threw it away.

All the fresh and dried natural ingredients were chosen based on a perceptible, existing aroma that was extracted and preserved through various processes, such as boiling in water. Or were they really? Aroma, seemingly invisible, was not captured as air, but rather through materials that could be seen and touched, which were transformed physically and chemically to smell stronger or different, and to look good together. What, then, was captured and preserved? Smell as a sign of life, existing or no longer living? Or smell as affecting ways of living in the present, past, or future? More simply, were existing scents actually captured, or were only new scents created through the mingling and merging of things? This confusion about the relationship between workshop processes and outcomes, and ultimately what smell even is and what conditions might support it, emphasized flourishing as something that is not possible to pause within an entangled capacity to materially and somatically affect and be affected. The smells did not last, and, as indicated by the resulting mold, life kept on growing.

DISCUSSION

In what follows, through the seasonal activities as examples, we reflect upon the ethics of wonder as a design aesthetics for planetary care. Our reflections emerged from a critique of the three seasonal activities in dialogue with one another. By returning to our understanding of aesthetics and drawing on a more-than-human perspective within design research, we considered how, for whom, or for what wonder was expressed and experienced before, during, and after the three activities.

The Prioritization of Human Experience

Related work highlights how "the tie of aesthetics to human experience may be limiting in addressing the full spectrum of design concerns" (Ikeya et al., 2025, p. 3). That is, a focus on human perception and the normative qualities of a "good" experience, such as delight, often excludes how more-than-humans encounter design and are affected by it. For example, research situated within biodesign calls for an aesthetic orientation toward "livingness" to foreground temporalities—often overlooked by people—to support multispecies flourishing (Karana et al., 2020). While the three seasonal activities presented do not explicitly delineate a staging of material relations between living beings, as seen through the inclusion of the wind, they all center on the impressions and questions of people.

In *Appreciating the Wind*, the FA's admiration of the pinwheel construction, in contrast to the disinterest of invited children, highlights differences in subjectivity across human participants. What was wonderful for an adult was not worth wondering about for some children. During the activity of *Looking for Frog Eggs*, bodily boundaries between species were questioned as people sought to fulfill their curiosity and complete a quest. This search to find a particular life form overshadowed the maintenance of boundaries that might be useful in supporting more-than-human forms of life. While in *Making Natural Perfumes*, flourishing was defined as what smells and looks good in the present to children and their parents. This grounded the encounter of unexpected flourishing as unwanted and not worthy of ongoing cultivation.

Across the three activities, we see a return to *how* bodily encounters are staged between people and more-than-human worlds as an important pathway to address the prioritization of human experience. *Appreciating the Wind* calls attention to which humans are invited to ask questions based on the form of the activities. That is, a failure to wonder might instead be a failure to include children in appropriate and meaningful ways. *Looking for Frog Eggs* suggests the finding of something else, such as unexpected life, to end searching and maintain not knowing. This supports keeping distances that might be best left unknown through the celebration of what is known. *Making Natural Perfumes* highlights the somatic mingling of bodies over detached observation based on established definitions. Thus, the focus shifts from qualifying flourishing as good or bad to questioning flourishing as *senseable* by people.

The Commodification of Nature

The prioritization of human needs, desires, and experiences exists within binary histories of classification that situate conceptions of "nature" as

something distinct from and in service of people (Haraway, 2003). In our work, through the framing of seasonal activities, nature is defined as more-than-human materials, beings, and forces in the outdoors for people to experience, use, and create with. Despite our intentions and those of the book to cultivate ecological curiosity and care *with* nature, the processes and outcomes of the three activities resemble familiar market products, and can commodify wonder itself as something to be consumed.

The desired and resulting outcome of *Appreciating the Wind* was a pinwheel, which shifted attention from bodily sensations of the wind as a force to its abstract representation through an object. Even if the wind was not quantified, it was measured, alongside the impression of being controllable through the ease of its human recreation. While *Looking for Frog Eggs* foregrounded searching and finding over the fishing nets as recognizable products, the activity objectified life as an identifiable body. Death also took the form of a body, reinforcing a dichotomy between life and death, and cause-and-effect relationships that are distant from human accountability. *Making Natural Perfumes* celebrated "natural" as shiny and artificial through a focus on the composition of ingredients, the shapes of bottles, and decorative elements. This assembling of a product positioned "being affected" as a hierarchical capturing of nature, containable through human perception.

From our critique of the three activities, we see a return to principles of *relationality* and *affection* as opportunities to address the commodification of nature amid the pursuit of wonder. *Appreciating the Wind* highlights the importance of abstract representations remaining close to the human body, through holding and running with the pinwheel, for example. Amid its absence, it supports possibilities of uncertainty alongside what a person feels, and makes present the effects and affects of people. *Looking for Frog Eggs* proposes taking time to admire differences across bodies. This shifts from categorizing life as absolute or distinct to celebrating ways of being regardless of their physical form. *Making Natural Perfumes* suggests attending to compositions of wonder as conditions for contradictions within that wonder. This recognizes that delight and disgust can thrive under the same conditions, and that coexistence might include a misalignment of presence.

CONCLUSION

In response to the ecological crisis, design calls for a new aesthetics that creates the conditions for other modes of relationality. We propose wonder as a possible design aesthetics through its ongoing questioning of coexistence, grounded within a staging of more-than-human bodily encounters and the experiencing of questions. From the presentation and

critique of three seasonal activities with children that situate wonder in design practice, we conclude with three implications for design. The first implication is the ongoing inclusion of children and the temporality of seasons in planetary care. This foregrounds pathways to design for the long-term and uncertain becoming of place-based relations and emergent flourishing. The second implication is supporting divergence from prescribed seasonal activities. This allows for a meaningful staging of bodily encounters between people of all ages and nature in daily life. The third implication is resisting the definition of how wonder should be experienced or expressed. This shifts a "successful" design aesthetics from a predictable form or unified outcome to a liminal space of affecting and being affected.

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